

OVERVIEW OF THE PROCESSING

Controller(s)

Field	Value
Controller	Operator of betting, market, prediction, trading, or sportsbook-style deployment
DPO or similar function	Controller DPO or privacy contact
Repository evidence	Consumers/BOOK_COM; betsClient.js; shybets manifest; patent/mainhub/dpia/ev

Name of the Processing

Internal name: Shyware Betting / Markets DPIA

SYSTEMATIC DESCRIPTION OF THE PROCESSING

High-level Description of the Processing

Processed Personal Data

#	Personal Data Item	Data Type / Data Subject Category	Sensitivity
1	Market/bet order	Event, side, stake, odds, timestamp, account commitment	Financial/
2	Eligibility/age/location status	Age, jurisdiction, geofence, exclusion status	High
3	Wallet/payment data	Account, payment, settlement, withdrawal metadata	Financial h
4	Risk and responsible-gaming data	Fraud, abuse, self-exclusion, limit signals	High

Purposes of the Processing Support market order entry, settlement, eligibility, responsible-gaming controls, fraud prevention, and audit.

ANALYSIS OF THE PROCESSING

Lawfulness and Minimisation

Customer controllers must assess gambling, market, payments, age-gating, geolocation, self-exclusion, and consumer-protection obligations. Shyware should not launch this domain without product-specific regulatory counsel.

Mainhub Patent Alignment

The Mainhub patent architecture can reduce canonical-state linkage between account commitments and market/bet records. It does not replace age verification, geolocation, self-exclusion, responsible-gaming monitoring, market-integrity records, payment controls, or gambling-law compliance. Betting deployments should not rely on canonical-state unlinkability as a substitute for regulated operator obligations.

Authority Partition This is a count-match embodiment. Four structurally separated parties are required:

Authority	Role	
Auth (Cognito)	Account layer; always required; gates access to IDV/KYC stage	S
IDV/KYC (Didit)	Verifies user identity on authenticated users	C
Market administrator	Determines domain eligibility; registered independently at scoping-identifier creation	C
Reconciling authority	Off-chain linkage store + sealer keys	I

Dual co-authorization enforcement (Claim 2): The validation layer must reject single-signature rescission transactions. **VERIFIED (run 26369476939):** *DUAL_COSIG section passes 379/379 — unauthenticated + valid token → 401; authenticated + no token → 400; both co-signatures → 200.*

CONSIDERATIONS ON NECESSITY AND PROPORTIONALITY

The betting and markets pattern may be proportionate only where the controller has a regulated basis for eligibility, payments, responsible-gaming controls, and market-integrity monitoring. Protected linkage and reveal workflows should be limited to legally required compliance, dispute, fraud, or user-requested recovery purposes. A controller should disable this domain until jurisdiction, age/location eligibility, payment flow, retention, and responsible-gaming duties are fixed.

RISK ASSESSMENT AND MANAGEMENT

#	Risk	Required Follow-up
1	Gambling/financial profiling	Minimize account-event linkage and retention
2	Ineligible user participation	Verify age, location, exclusion, and jurisdiction controls
3	Harmful behavior detection	Define responsible-gaming signals and controller duties
4	Payment/settlement exposure	Assess payment processor role and transfer posture

DPIA Audit Findings

This DPIA is the audit record for betting and market deployments. Before production, the DPIA must identify the provider partition across age/location verification, payment or wallet provider, protected linkage store, reconciling authority, canonical commit authority, and telemetry/support providers. Canonical-state unlinkability does not reduce the need for gambling-law compliance, responsible-gaming monitoring, sanctions/payment controls, or market-integrity records. The software evidence for the core kernel may be complete while the controller still needs to fix jurisdiction, eligibility, recovery/reveal posture, retention, and payment settlement flows for this domain.

Operator Compliance Instructions For the non-code governance residuals identified in this DPIA, the operator is instructed by the **Shyware SDK Compliance Guide** (`patent/mainhub/dpia/source/COM` published at <https://shyware.fyi/legal>). Key operator obligations:

- **Art. 6 lawfulness basis** — Identify and document the legal basis for each processing purpose before production.
- **Art. 28 sub-processor DPAs** — Execute signed DPAs with all subprocessors listed in the generated Records of Processing (`node scripts/generate-rop.mjs -config shyconfig.json`).
- **Art. 30 Records of Processing** — Generate from shyconfig using `generate-rop.mjs`; complete all `REQUIRED_OPERATOR_INPUT` fields; maintain under Art. 30(1).
- **Storage limitation (Art. 5(1)(e))** — Add `retention` block to `shyconfig.json`; configure and operationally enforce the `deletion_cron` schedule.
- **Three-authority key independence** — Verify at scoping-identifier creation that `eligibility_authority_key` $\neq$ `reconciling_authority_key` $\neq$ `auth_key`.
- **Quarterly independent audit** — Engage an independent auditor to verify three-authority operational separation before production.

Residual Risk Assessment

The residual scores below assume the core Shyware twin-kernel controls are implemented and evidence-linked. The software evidence for the core kernel may be complete while betting/market regulatory, responsible-gaming, payment, eligibility, or operator-policy choices still need to be fixed for this domain. These scores describe residual domain risk after validation; they do not say the kernel tests were missing.

Risk	Additional Measures Applied
Gambling or financial profiling	Account-event minimization, retention limits, and restricted analytics
Ineligible user participation	Age, location, exclusion, and jurisdiction checks
Responsible-gaming failure	Harm signals, limits, self-exclusion, and controller intervention process

Risk	Additional Measures Applied
Payment or settlement exposure	Payment processor review, tokenization, and transfer controls
Provider consolidation	Provider matrix for eligibility, payment/wallet, linkage, reconcile, commit, and

INVOLVEMENT OF INTERESTED PARTIES

Privacy counsel, gambling or market regulatory counsel, payment/compliance stakeholders, security, and responsible-gaming stakeholders should review this DPIA before production. Data subject or representative consultation may be appropriate where self-exclusion, vulnerable users, or high-risk consumer-protection duties apply.

CONCLUSION AND DECISION

Draft decision: Separate DPIA and gambling/market regulatory review required before production.